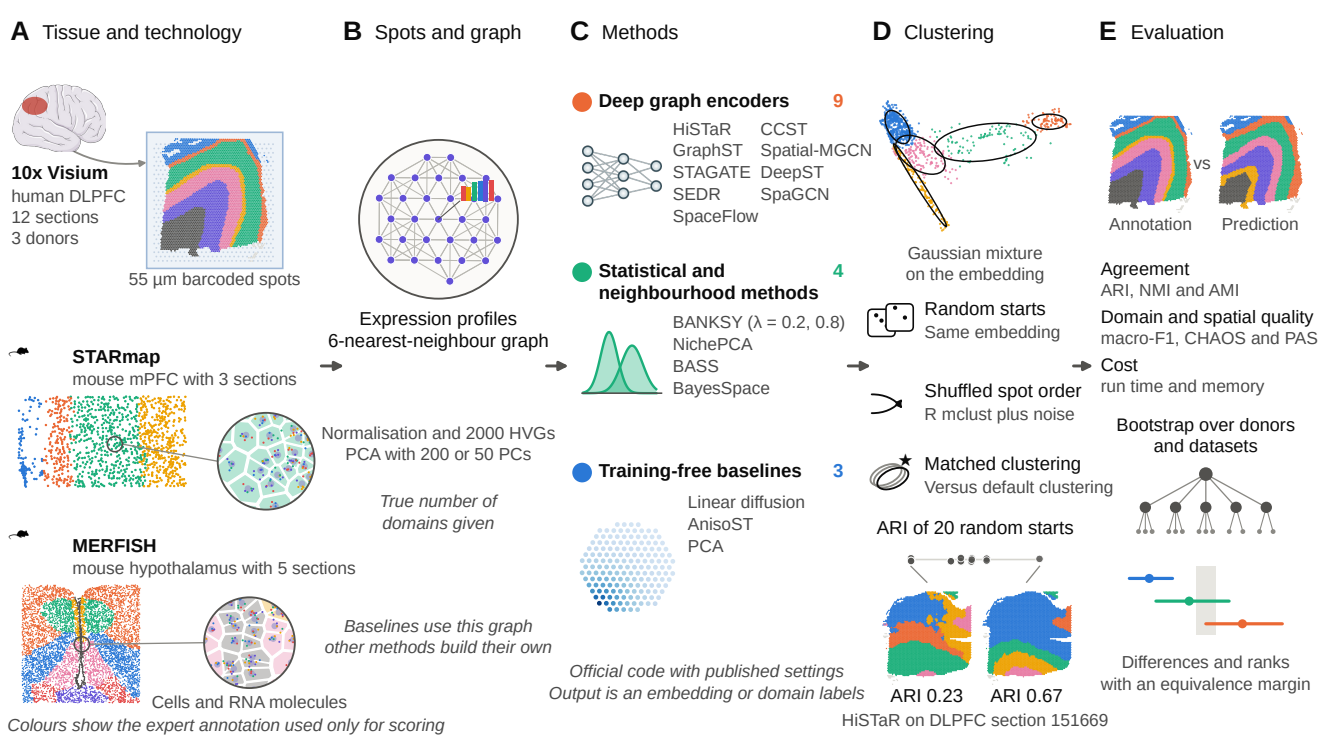

A Tissue and technology
B Spots and graph
C Methods
D Clustering
E Evaluation
10x Visium
human DLPFC
12 sections
3 donors
55 µm barcoded spots
Expression profiles
6-nearest-neighbour graph
Deep graph encoders 9
HiSTaR
GraphST
STAGATE
SEDR
SpaceFlow
CCST
Spatial-MGCN
DeepST
SpaGCN
Statistical and neighbourhood methods 4
BANKSY (λ = 0.2, 0.8)
NichePCA
BASS
BayesSpace
Training-free baselines 3
Linear diffusion
AnisoST
PCA
Official code with published settings
Output is an embedding or domain labels
Gaussian mixture on the embedding
Random starts
Same embedding
Shuffled spot order
R mclust plus noise
Matched clustering
Versus default clustering
ARI of 20 random starts
ARI 0.23
ARI 0.67
HiSTaR on DLPFC section 151669
vs
Annotation
Prediction
Agreement
ARI, NMI and AMI
Domain and spatial quality
macro-F1, CHAOS and PAS
Cost
run time and memory
Bootstrap over donors and datasets
Differences and ranks with an equivalence margin
STARmap
mouse mPFC with 3 sections
Normalisation and 2000 HVGs
PCA with 200 or 50 PCs
True number of domains given
MERFISH
mouse hypothalamus with 5 sections
Cells and RNA molecules
Baselines use this graph other methods build their own
Colours show the expert annotation used only for scoring